

AirNav Risk Analysis Manual

A practitioner guide to twelve safety risk assessment methods for air navigation service provision

- Bowtie
- JHA
- FHA
- Fatigue modelling
- HAZID
- FMEA / FMECA
- STPA
- FRAM
- HAZOP
- LOPA
- FTA
- Bayesian Belief Networks

Document control

Item	Detail
Title	AirNav Risk Analysis Manual
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Ed.	Date	Description	Prepared	Approved
0.1	Sep 2026	First draft: framework and 12 method chapters	SQRM drafting team	—

How to use this manual

Part A sets out the common framework that every AirNav safety risk assessment must follow: the risk management process, the risk classification scheme, how to choose a method and how to run a workshop. Part B contains one chapter per method. Every method chapter follows the same pattern — *at a glance*, key concepts, step-by-step procedure, worksheet template, a worked ANSP example, tips and pitfalls, and key references — so that an assessor can pick up any chapter on its own. Part C explains how to combine outputs into a Safety Assessment Report and keep them alive in the hazard log.

All numbers in the worked examples (failure rates, probabilities, frequencies, conditional probability tables) are **illustrative**. They were chosen to demonstrate the arithmetic and must not be quoted as AirNav data. Real assessments must use AirNav occurrence data, equipment reliability records, manufacturer data or documented expert judgement, and state their source.

Words such as **shall** indicate a requirement of this manual; **should** indicates recommended practice; **may** indicates an option.

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1. Introduction

1.1 Purpose

An air navigation service provider (ANSP) manages risk every time it changes a system, a procedure, its airspace or its organisation, and every time it learns from an occurrence. ICAO Annex 19 requires service providers to implement a Safety Management System (SMS) in which hazards are identified and the associated safety risks are assessed and mitigated. ICAO does not prescribe which analytical method to use; the choice is left to the service provider, provided the method is appropriate to the nature and complexity of the change or the question being asked.

This manual gives AirNav staff a single, consistent reference for twelve methods that are either widely used in ATM or that bring a capability the classic tools lack. It is written for safety specialists, facilitators and the operational and technical subject-matter experts who take part in assessments.

1.2 Regulatory and guidance basis

Source	Relevance to this manual
ICAO Annex 19 — Safety Management [R1]	Framework of four SMS components; hazard identification and safety risk assessment and mitigation are mandatory elements of the Safety Risk Management (SRM) component.
ICAO Doc 9859 — Safety Management Manual, 4th ed. 2018 [R2]	Guidance on hazard identification, safety risk probability and severity, the risk matrix and tolerability, and management of change. Basis of Chapter 3.
CASR Part 19 — Safety Management System, issued by Minister of Transportation Regulation PM 2 Tahun 2026 [R3]	National SMS requirement for Indonesian aviation service providers, structured around safety policy and objectives, safety risk management, safety assurance and safety promotion.
ICAO Annex 11 and Doc 9966 [R4, R5]; ICAO/CANSO/IFATCA Fatigue Management Guide for ATS Providers [R6]	Requirements and guidance for managing ATC fatigue by prescriptive limits and/or a Fatigue Risk Management System. Basis of Chapter 15.
EUROCONTROL SAM and ESARR 4 [R7, R8]	Widely used ATM safety assessment lifecycle (FHA → PSSA → SSA) and ATM severity classification; good practice reference for Chapter 12.
FAA ATO SMS Manual and SRM Guidance [R9]	ANSP example of an SRM process that uses a tool-box of hazard analysis methods.
ISO 31000:2018 and IEC 31010:2019 [R10, R11]	Generic risk management principles and the catalogue of risk assessment techniques.
Method standards [R12-R36]	Bowtie (UK CAA), HAZID (ISO 17776), HAZOP (IEC 61882), JHA (OSHA 3071), FMEA/FMECA (IEC 60812), LOPA (CCPS), FHA (SAE ARP4761A, EUROCONTROL SAM), STPA (Leveson & Thomas), FTA (IEC 61025), fatigue models, FRAM (Hollnagel), BBN (Fenton & Neil).

1.3 Key definitions

Term	Definition used in this manual
Hazard	A condition or an object with the potential to cause or contribute to an aircraft incident or accident (ICAO). Example: “aircraft and vehicles moving on or near an active runway”.
Safety risk	The predicted probability and severity of the consequences or outcomes of a hazard (ICAO).
Consequence / outcome	The potential end state of a hazard, e.g. runway collision, mid-air collision, controlled flight into terrain.
Risk control (barrier, mitigation, safeguard)	A measure that reduces the probability and/or the severity of a consequence. Different methods use different names: barrier (bowtie), safeguard (HAZOP), independent protection layer (LOPA), control / safety constraint (STPA).
Safety objective / safety requirement	A statement of the maximum tolerable frequency of a hazard (objective) and the means to achieve it (requirement), used in the FHA/PSSA lifecycle.
ALARP	As low as reasonably practicable: a tolerable risk must be reduced further unless the cost of doing so is grossly disproportionate to the benefit.
Hazard log	The controlled register of identified hazards, their assessed risk, controls, owners and status (see Part C).

2. The safety risk management process

Every assessment, whatever method is used, follows the six-step process in Figure 1. The methods in Part B are tools that support steps 2 to 5; none of them replaces the process.

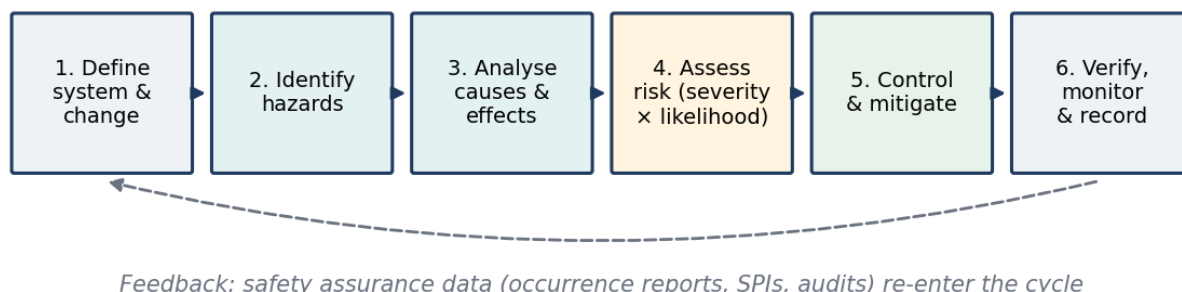


Figure 1. AirNav safety risk management process (after ICAO Doc 9859 and ISO 31000).

2.1 When a safety assessment is required

- Introduction or modification of ATM, CNS or supporting systems (e.g. new ATM automation, ADS-B, VCCS, AMHS, power systems).
- New or changed ATS procedures, airspace structure, routes, instrument flight procedures, sector configuration or opening schemes.
- Organisational changes affecting operations: manning levels, rosters and shift patterns, relocation of units, contracting of services, training arrangements.
- Temporary changes: works on the manoeuvring area, degraded modes, contingency operations, special events.
- After a serious incident or a trend in safety performance indicators (SPIs) that suggests a control is not working.

2.2 Roles

Role	Responsibilities
Change sponsor / system owner	Describes the change, provides design and operational information, owns the resulting safety requirements and actions.
Safety assessor / facilitator	Selects the method(s), prepares and runs the sessions, records results, challenges assumptions, drafts the Safety Assessment Report.
Subject-matter experts (SMEs)	Current operational ATCOs, supervisors, ATSEP/engineers, AIM and procedure designers, and where relevant pilots, airport operator and military. They bring work-as-done knowledge.
Safety Manager	Assures the quality and consistency of assessments, maintains the hazard log, checks that the risk classification scheme is applied correctly.
Risk acceptance authority	Accepts residual risk at the level set in Table 3.3, up to the Accountable Executive for the highest tolerable risks.

2.3 Describing the system

The single most common cause of a weak assessment is an unclear scope. Before any method is applied, the assessor shall document: the purpose of the change; the system boundary (people, procedures, equipment, environment and interfaces — the “SHELL” elements); the operational environment (airspace class, traffic volume and mix, weather, time of day); the assumptions made; and the before/after comparison where the change modifies an existing service.

3. Risk classification scheme

AirNav uses a 5 × 5 qualitative matrix based on the example in ICAO Doc 9859 [R2]. Severity is always assessed for the **most credible** outcome, not the worst conceivable one; where two outcomes are both credible (for example a minor loss of separation that is likely and a mid-air collision that is remote), both should be assessed and the higher risk index retained.

3.1 Likelihood

Level	Descriptor	Qualitative meaning	Indicative quantitative band (per operating hour of the affected unit/system)*
5	Frequent	Likely to occur many times; has occurred frequently	$> 10^{-3}$
4	Occasional	Likely to occur sometimes; has occurred infrequently	10^{-5} to 10^{-3}
3	Remote	Unlikely to occur, but possible; has occurred rarely	10^{-7} to 10^{-5}
2	Improbable	Very unlikely to occur; not known to have occurred	10^{-9} to 10^{-7}
1	Extremely improbable	Almost inconceivable that the event will occur	$< 10^{-9}$

Calibration. * The quantitative bands are an AirNav proposal to allow quantitative methods (FTA, LOPA, BBN) to be mapped onto the matrix. They must be calibrated against AirNav traffic exposure and approved by the Safety Review Board before use. ICAO Doc 9859 gives only qualitative descriptors.

3.2 Severity

Level	Descriptor	ATM meaning (examples)
A	Catastrophic	Accident: mid-air collision, runway collision, CFIT; multiple fatalities; equipment destroyed.
B	Hazardous	Serious incident: large reduction in safety margins, e.g. separation lost with high collision risk (ACAS RA required to avoid collision), runway incursion with narrowly avoided collision; serious injury.
C	Major	Major incident: significant reduction in safety margins, e.g. separation lost but controlled recovery; ATCO or crew workload significantly increased; total loss of a service with fallback in place.
D	Minor	Significant incident: slight reduction in margins, e.g. separation infringement with no collision risk, nuisance alerts, minor operating limitations.
E	Negligible	No immediate effect on safety; little consequence.

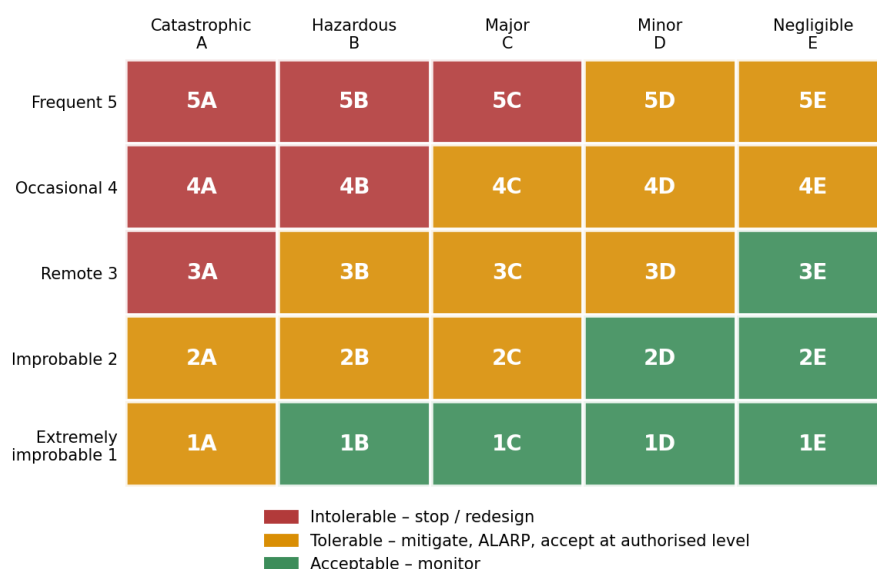


Figure 2. AirNav safety risk matrix and tolerability regions (adapted from the ICAO Doc 9859 example).

3.3 Tolerability and acceptance authority

Region	Risk indices	Required action	Acceptance authority
Intolerable	5A, 5B, 5C, 4A, 4B, 3A	Do not introduce the change or continue the operation. Redesign or add controls until the risk moves into the tolerable region.	Cannot be accepted
Tolerable (upper)	5D, 4C, 3B, 2A	Additional controls required; demonstrate ALARP; risk may be accepted only with a documented justification and monitoring plan.	Accountable Executive (President Director) on advice of Safety Review Board
Tolerable (lower)	5E, 4D, 4E, 3C, 3D, 2B, 2C, 1A	Mitigate where reasonably practicable; document ALARP argument.	Director of Operations / Director of Engineering (as applicable)
Acceptable	3E, 2D, 2E, 1B, 1C, 1D, 1E	Accept; monitor through normal safety assurance.	Unit / branch manager

To confirm. The split of the tolerable region and the named authorities are proposals for AirNav to confirm; they must match the delegations in the AirNav SMS Manual.

3.4 Quantitative safety targets

Quantitative methods (FTA, LOPA, BBN, FMECA) produce frequencies or probabilities. They must be compared with a quantitative target that is consistent with the matrix above. A well-known ATM reference is ESARR 4 [R8], which set the maximum tolerable probability of ATM *directly contributing* to an accident at 1.55×10^{-8} per flight hour for commercial air transport in the ECAC area. That value was derived from European accident statistics and traffic; AirNav should derive its own targets from its accident and traffic data, apportioned across hazards, rather than import the European figure unchanged.

Two rules apply to all quantitative work: state the unit (per flight hour, per operation, per sector hour, per year) and never compare numbers with different units; and treat results as order-of-magnitude estimates — a factor of 3 is within the uncertainty of most inputs.

4. Selecting a method

No single method answers every safety question. The methods differ in what they model (functions, components, tasks, control loops, everyday variability), in whether they quantify, and in when they are most useful in the life of a change. Table 4.1 summarises the twelve methods; Figure 3 maps them onto the lifecycle.

Method	Main question it answers	Type	Typical ANSP use	Effort
Bowtie	What prevents this top event, what limits its consequences, and are those barriers healthy?	Qualitative, barrier-based	Communicating major hazards (runway incursion, airborne conflict, CFIT); SMS risk register; monitoring barrier health	Medium
HAZID	What hazards could this change introduce?	Qualitative, guideword brainstorm	Early concept of a change, new unit, airspace redesign	Low
HAZOP	How could the design or procedure deviate from its intent, and with what effect?	Qualitative, systematic guidewords	Detailed procedures, coordination and data flows (AIDC/OLDI, CPDLC, AMHS)	High
JHA	What can hurt the person doing this task, step by step?	Qualitative, task-based	Engineering maintenance, work at height, RF/electrical work, airside vehicle tasks	Low
FMEA / FMECA	How can each component fail and what does each failure do to the service?	Semi-quantitative, bottom-up	CNS/ATM equipment, power, networks; maintenance strategy	Medium-High
LOPA	Is one scenario controlled by enough independent layers to meet the target?	Semi-quantitative, scenario-based	Checking barriers against a frequency target; justifying a new barrier	Medium
FHA	What happens if a function is lost, degraded or erroneous, how severe is it, and how often may it happen?	Qualitative + safety objectives	Start of the SAM-style lifecycle for systems and services	Medium
STPA	How could the control structure (people, automation, procedures) issue unsafe control actions?	Qualitative, systems-theoretic	Automation, new concepts, human-automation interaction; CAST for investigations	High
FTA	Which combinations of basic events cause this top event, and how likely is it?	Quantitative, top-down deductive	Loss of surveillance/communications; proving safety objectives in PSSA/SSA	Medium-High
Fatigue modelling	How alert will controllers be across this roster?	Quantitative, bio-mathematical	Roster design, FRMS, change of shift pattern, investigations	Low-Medium
FRAM	How does everyday performance variability combine to produce success or failure?	Qualitative, functional	Understanding complex work-as-done; Safety-II; investigations	High
BBN	How do multiple uncertain factors combine, and what do observations tell us about causes?	Quantitative, probabilistic graph	Integrated risk models, SPI analysis, human and organisational factors	High

	Concept / change proposal	Design / specification	Implement. & transition	Operations	Occurrence investigation	Safety performance monitoring
Bowtie	○	●	○	●	○	●
HAZID	●	○	○	○		
HAZOP	○	●	○	○		
JHA			●	●	○	
FMEA/FMECA		●	○	○	○	
LOPA		○	○	○	○	○
FHA	●	●	○			
STPA	●	●	○	○	●	
FTA		●	○	○	●	○
Fatigue model	○		○	●	●	●
FRAM	○	○	○	●	●	○
BBN	○	○		○	●	●

Figure 3. Where each method adds most value in the life of a change (● primary use, ○ supporting use).

4.1 Recommended combinations

Situation	Recommended chain of methods
New or modified CNS/ATM system	HAZID or FHA to set hazards and safety objectives → FMEA of the architecture → FTA to show objectives are met (with common-cause analysis) → bowtie update for the affected major hazards.
New or changed ATS procedure / airspace	HAZID → HAZOP of the procedure steps and coordination → STPA if automation or new controller-pilot interactions are involved → bowtie; LOPA for any scenario that relies on few barriers.
New automation or controller tools (e.g. AMAN, CPDLC, conflict probes)	FHA → STPA (unsafe control actions, mode confusion) → FTA for quantitative objectives.
Roster or staffing change	Fatigue modelling of the old and new rosters → JHA-style task review of critical periods → bowtie barrier review (supervision, relief, STCA).
Engineering maintenance task	JHA, with FMEA input on the effect on service if the maintained equipment is taken out of service.
Occurrence investigation or recurring trend	FRAM or CAST (STPA for accidents) to understand work-as-done → FTA for technical failures → BBN to combine evidence → bowtie to show which barriers failed.
Organisation-wide risk picture	Bowtie library for top hazards → BBN linking SPIs and precursors to the bowtie barriers.

5. Conducting an assessment workshop

Most methods in this manual rely on a structured team session. The quality of the result depends far more on who is in the room and how the session is run than on the worksheet.

5.1 Preparation

- Agree the scope, system description and assumptions with the change sponsor in writing before the session.
- Choose participants: 4–8 people is ideal. Include at least one current front-line practitioner for each affected role (ATCO, supervisor, ATSEP) and, where the change affects them, airspace users and the airport operator.
- Send a pre-read (system description, drawings, draft procedures, previous assessments and relevant occurrences) at least three working days in advance.
- Prepare the worksheet template and guidewords for the chosen method; pre-populate what can be pre-populated (functions, nodes, task steps) so the session time is spent on analysis.

5.2 Facilitation

- Open with the purpose, the scope boundary, the risk classification scheme and the rule that every view is recorded.
- Keep the facilitator and the scribe as separate people; the scribe records in the tool or worksheet in view of the team.
- Watch for known biases: *anchoring* on the first estimate, *optimism* about new systems, *groupthink* and deference to seniority. Ask the most junior practitioner first when estimating likelihood.
- Distinguish controls that exist and are verified from controls that are planned or assumed. Only verified controls may be credited when estimating current risk.
- Limit sessions to about three hours; long sessions degrade quality.

5.3 Evidence and recording

For each likelihood and severity judgement, record the basis: occurrence data (with period and exposure), reliability data, analogous systems, or expert judgement (with the names or roles of those who judged). Record dissenting views. Every action shall have an owner and a due date, and every hazard shall be entered into the hazard log with a unique identifier.

6. Bowtie analysis

At a glance	
Purpose	Show, for one hazard and one top event, all the threats that can release it, the consequences that can follow, and the barriers on each path; then manage the health of those barriers.
When to use	Major operational hazards (runway incursion, airborne conflict, CFIT, loss of control, runway excursion, wake encounter); SMS risk register; communicating risk to management and staff; after an occurrence to show which barriers failed.
Inputs	Hazard and top-event definition; occurrence data; procedures and system descriptions; existing HAZID/HAZOP results. The UK CAA bowtie templates are a ready starting point [R12].
Outputs	Bowtie diagram; barrier register with type, owner, effectiveness and criticality; list of weak or missing barriers; links to SPIs.
Team and time	Facilitator + 4-6 SMEs; one to two half-day sessions per bowtie.
Key references	UK CAA bowtie guidance and CAP1329 [R12]; IEC 31010:2019 [R11]; CCPS/Energy Institute bowtie guidelines.

6.1 Key concepts

The bowtie joins a fault-tree-like left side (causes) and an event-tree-like right side (consequences) around a central **top event** — the moment control over the hazard is lost. Its strength is that it is easy to read and puts the emphasis on the **barriers** that an organisation must keep working.

Element	Meaning	ANSP example
Hazard	Normal part of the operation that has potential for harm.	Aircraft and vehicles moving on or near an active runway.
Top event	The release of the hazard; loss of control over it. Must be before damage occurs.	Unauthorised entry of an aircraft, vehicle or person onto the active runway.
Threat	A possible direct cause that can release the top event on its own.	Pilot misinterprets taxi clearance.
Consequence	Undesired outcome resulting from the top event.	Runway collision.
Prevention barrier	Stops a threat from producing the top event.	Stop bars operated H24; standard phraseology with read-back.
Recovery (mitigation) barrier	Stops the top event from developing into the consequence, or reduces severity.	A-SMGCS/RIMCAS alert and ATCO intervention.
Escalation factor	A condition that degrades a barrier.	Stop bar unserviceable; low visibility; ATCO fatigue.
Escalation-factor barrier	Keeps the escalation factor from defeating the main barrier.	Procedure for stop-bar U/S: conditional clearances prohibited, follow-me mandatory.

A barrier must be able, on its own, to stop the sequence on that path. A training course, a policy or “awareness” is usually not a barrier by itself — it is a support to a barrier (for example, RT training supports the barrier “read-back / hear-back”). Recording supports as escalation-factor barriers keeps the diagram honest.

6.2 Procedure

1. **Define the hazard and top event.** One top event per bowtie. Phrase it as a loss of control, not as damage.
2. **Identify threats.** Use occurrence data, HAZID output and the CAA template threats. Each threat must be sufficient on its own to cause the top event.
3. **Identify consequences.** Usually three to six credible outcomes; assess each with the risk matrix (Chapter 3).
4. **Identify prevention and recovery barriers** for each path, in time order from threat to consequence. Test each against the barrier criteria: effective, independent of the threat and other barriers on the path, and auditable.
5. **Add escalation factors and their barriers** for barriers that are known to degrade.
6. **Characterise each barrier:** type (hardware, software, human action, human-hardware), owner, effectiveness rating and criticality, and link it to an SPI or audit item.
7. **Review** the diagram with the team, identify weak paths (few or poor barriers) and raise actions.
8. **Keep it alive:** update after occurrences, audits and changes; use it to code occurrence reports by barrier failed [R12].

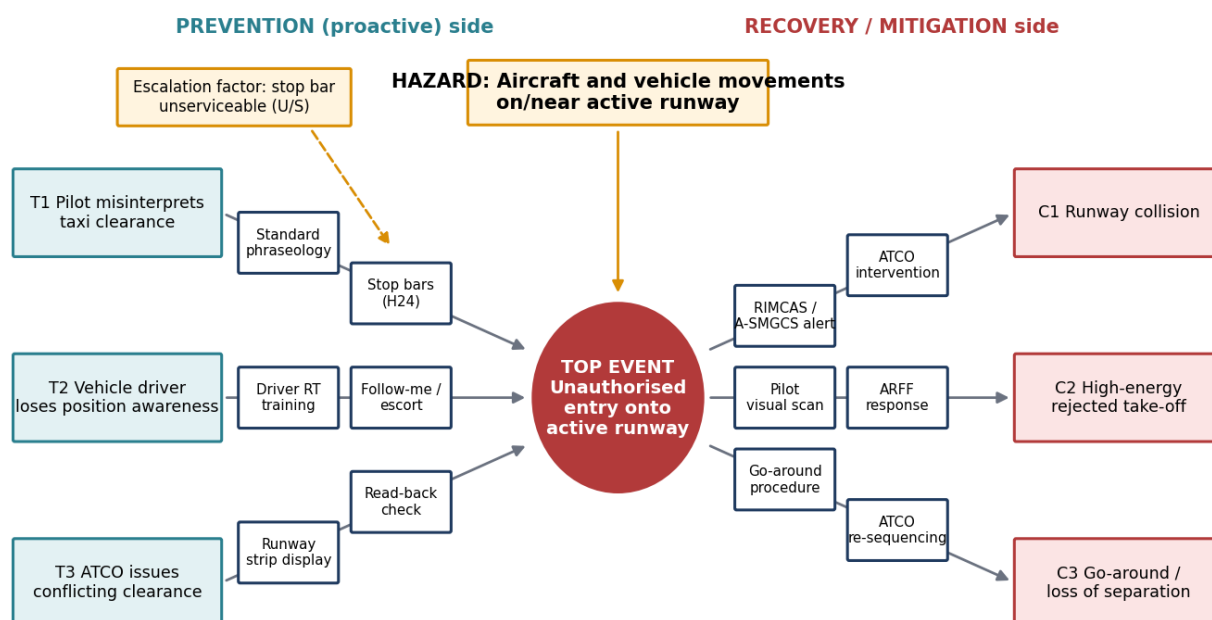


Figure 4. Simplified bowtie for unauthorised entry onto an active runway (illustrative; one escalation factor shown).

6.3 Barrier register template

ID	Barrier	Side / path	Type	Owner	Effect.	Crit.?	Monitoring (SPI/audit)
PB-02	Stop bars operated H24 at all runway-holding positions	Prevention / T1	Hardware + human	Aerodrome operator; TWR supervisor	Good	Yes	Stop-bar serviceability %; stop-bar crossing reports
PB-05	Read-back / hear-back of runway clearances	Prevention / T1, T3	Human	TWR unit chief	Good	Yes	RT sampling audits; read-back error rate
RB-01	RIMCAS / A-SMGCS runway incursion alert	Recovery / all	Hardware + software	CNS engineering	Good (poor in heavy rain clutter)	Yes	Alert availability; false/nuisance alert rate

ID	Barrier	Side / path	Type	Owner	Effect.	Crit.?	Monitoring (SPI/audit)
RB-02	ATCO intervention (stop / go-around instruction)	Recovery / all	Human	TWR unit chief	Good	Yes	Proficiency checks; simulator scenario results
EF-01	Stop bar U/S procedure (no conditional clearances, follow-me)	EF barrier on PB-02	Procedure	Aerodrome ops	Poor (not always applied)	Yes	Audit of NOTAM/U/S logs vs clearances

6.4 Tips and pitfalls

- Do not draw a top event that is already a consequence (“runway collision” is a consequence, not a top event).
- Do not list the same barrier on every line to make the diagram look strong; ask whether it is truly independent of the threat.
- Effectiveness ratings should be based on evidence (audits, occurrence data), not on the designer’s intent.
- Start from the UK CAA bowtie templates (the library listed at caa.co.uk includes airborne conflict, CFIT, runway excursion, runway incursion, ground handling, fire and loss of control) and adapt them to AirNav’s operation rather than starting from a blank page.

7. HAZID — Hazard Identification study

At a glance	
Purpose	Identify, early and broadly, the hazards that a proposed change or operation could introduce, using a guideword checklist to prompt a structured brainstorm.
When to use	Concept and planning stage of any change; new units or facilities; airspace or route redesign; transition plans. Often the first safety activity of a project.
Inputs	Concept of operations; scope and boundary; list of stakeholders; historic occurrences in similar operations.
Outputs	HAZID register: hazards, causes, consequences, existing controls, initial risk index, actions; list of issues needing a more detailed method.
Team and time	Facilitator, scribe, 5–10 SMEs from all affected disciplines; half a day to two days.
Key references	ISO 17776:2016 [R13]; IEC 31010:2019 [R11]; ICAO Doc 9859 hazard identification [R2].

7.1 Key concepts

HAZID works at a higher level than HAZOP. Instead of examining every step of a design, the team walks through a list of **guideword categories** and asks, for each: “could this be a source of hazard for this change?”. Its value lies in breadth — making sure no category is forgotten — and in bringing different disciplines together early, when design changes are still cheap.

7.2 ANSP guideword checklist

Category	Prompt guidewords
People	Staffing levels; competence and rating; training for the change; workload peaks; fatigue; language proficiency; supervision; handover/takeover
Procedures	New or changed procedures; degraded modes; contingency; coordination with adjacent units; phraseology; letters of agreement
Equipment / systems	CNS coverage; ATM system functions; HMI; alerts (STCA, MSAW, APW, RIMCAS); data quality; cyber security; power; fallback
Airspace and aerodrome	Route structure; sectorisation; restricted/danger areas; terrain; MSA; runway configuration; hot spots; works in progress
Traffic	Volume and complexity; mix (IFR/VFR, military, helicopters, drones); crossing flows; special operations
Environment	Weather (convective, low visibility, volcanic ash); night; seasonal peaks; bird activity
Interfaces	Adjacent ACC/APP/FIRs; airport operator; airlines; military; MET; AIM; external suppliers
Transition	Cut-over; parallel running; reversion; training during live operations; data migration
Organisation	Roles and responsibilities; contracting; management of change; reporting culture

7.3 Procedure

1. Brief the team on the change, scope and assumptions; agree the guideword list (tailor Table 7.2).
2. For each guideword, brainstorm hazards (“what could go wrong?”), their causes and consequences. Record everything, filter later.
3. For each hazard, record the existing controls and make an initial risk assessment with the AirNav matrix.
4. Identify further controls and actions; flag items needing a detailed method (HAZOP, FTA, STPA, LOPA).
5. Review the register for duplicates, merge, and transfer hazards to the hazard log.

7.4 Worked example — relocation of an approach unit to a new ATC centre

ID	Guideword	Hazard	Cause(s)	Consequence	Existing controls	Risk	Action
HZ-01	Transition	Loss of surveillance display at cut-over	Data feed not re-routed; configuration error	Loss of separation during cut-over	Cut-over at low traffic; old site on hot standby	3C	HAZOP of cut-over plan; rollback criteria
HZ-02	People	ATCOs unfamiliar with new CWP HMI	Short training window	Delayed or erroneous inputs	Simulator training	4D	Minimum hours on new CWP before live ops
HZ-03	Interfaces	Wrong telephone coordination lines to TWR	VCCS mapping error	Late coordination of missed approach	Test plan	3C	End-to-end line check with each TWR
HZ-04	Equipment	Single power feed to new building	Design	Total loss of service	UPS 30 min	2B	Second feed or genset; FMEA of power
HZ-05	Environment	New site outside radio line of sight of low-level sector	Topography	Loss of VHF contact below FL050	None yet	3B	Coverage study; remote radio site

7.5 Tips and pitfalls

- HAZID is not the end of the assessment: its output is a list of hazards, many of which need a detailed method.
- Resist debating risk ratings at length in a HAZID; a provisional rating is enough to prioritise.
- Record hazards neutrally (“loss of ...”, “incorrect ...”), not as accusations.

8. HAZOP — Hazard and Operability study

At a glance	
Purpose	Systematically examine a defined design or procedure, part by part, for deviations from the design intent, using standard guidewords applied to parameters.
When to use	Detailed design of procedures, coordination processes and data flows (e.g. AIDC/OLDI messages, CPDLC dialogues, flight plan processing), and of technical systems when drawings are mature.
Inputs	Frozen design or draft procedure; flow diagrams, message sets, sequence charts; operating assumptions.
Outputs	HAZOP worksheets per node; recommendations; list of operability issues.
Team and time	Independent chair, scribe, designer, operator (ATCO/ATSEP), specialist as needed; typically 3-6 nodes per day.
Key references	IEC 61882:2016 Hazard and operability studies — application guide [R14]; IEC 31010:2019 [R11].

8.1 Key concepts

The system is divided into **nodes** (a procedure step, a message exchange, a system element). For each node the team states the **design intent**, identifies the **parameters** (e.g. message content, timing, sequence, flow of information, action) and combines each parameter with each **guideword** to generate a **deviation**. For each meaningful deviation, the team records causes, consequences, existing safeguards and recommendations.

Guideword	Meaning	ATM interpretation (examples)
No / Not	Complete negation of intent	Message not sent; clearance not issued; coordination not done
More	Quantitative increase	More aircraft than planned; higher level than coordinated; message sent twice
Less	Quantitative decrease	Less information; lower level; shorter spacing
As well as	Qualitative increase	Correct message plus spurious one; clearance plus unintended instruction
Part of	Qualitative decrease	Incomplete message (level without conditions); partial read-back
Reverse	Logical opposite	Climb instead of descend; transfer in wrong direction
Other than	Complete substitution	Message sent to wrong unit; clearance to wrong callsign
Early / Late	Timing relative to clock time	Estimate sent too late for acceptance; frequency change too early
Before / After	Order or sequence	Transfer of control before coordination accepted

8.2 Procedure

1. Define scope and objectives; assemble the team; collect design documents.
2. Divide the design into nodes and state the design intent of each.
3. For each node, apply each relevant parameter-guideword pair to generate deviations.
4. For each credible deviation: identify causes, consequences (assume safeguards fail to see the unmitigated outcome), existing safeguards, risk index and recommendations.
5. Record as you go; the chair confirms each row before moving on.
6. Follow up recommendations to closure and feed hazards into the hazard log.

8.3 Worked example — ATS inter-centre data coordination (estimate message)

Node 3: “Upstream ACC sends estimate (EST/CPL) message to downstream ACC at least N minutes before the transfer point; downstream system displays the flight in the pending list; downstream ATCO accepts”. Design intent: the downstream sector receives complete, correct and timely data for every flight crossing the boundary.

Parameter + guideword	Deviation	Possible causes	Consequences	Safeguards	Risk	Recommendation
Message + NO	Estimate not received	Link failure; flight plan not activated; address error	Unknown traffic enters sector; late or no coordination	Link-status alarm; verbal coordination backup	3C	Alert downstream ATCO when a flight is within X min of boundary with no EST
Level + OTHER THAN	Coordinated level differs from actual	ATCO changes level after EST sent; no revision (ACP/REV)	Level conflict in downstream sector	Mode C/S check at transfer; STCA	3B	Automatic revision message on cleared-level change; CLAM
Timing + LATE	Estimate arrives inside the agreed time	Short-notice route change; queue delays	Downstream cannot plan; workload	LoA fallback: verbal	4D	Monitor late-EST rate as SPI
Message + AS WELL AS	Duplicate flight created	Re-sent message not correlated	Two labels; wrong one used	Code/callsign correlation	3C	Duplicate-detection logic test
Sequence + BEFORE	Transfer of communication before acceptance	Workload; habit	Aircraft calls unprepared sector	Procedures; training	4D	System inhibits transfer before acceptance

8.4 Tips and pitfalls

- HAZOP needs a mature design. Applied too early it degenerates into a HAZID.
- The chair must be independent of the design team and keep the pace steady; fatigue sets in after about four hours.
- Human-action nodes (procedural HAZOP) work well with the timing and sequence guidewords; the team must include a current practitioner.

9. JHA — Job Hazard Analysis

At a glance	
Purpose	Protect the people who carry out a task by breaking the job into steps, identifying the hazards of each step and specifying controls.
When to use	Engineering and maintenance tasks (radar, NDB/DVOR/DME, ILS, VHF sites, towers, generators, UPS, HVAC), airside work, vehicle operations, any new, infrequent or high-risk task, and after an injury or near miss.
Inputs	Task description, manufacturer manuals, site survey, previous incidents, legal OHS requirements.
Outputs	JHA sheet per job: steps, hazards, controls, responsible person; input to work instructions, permits to work and toolbox talks.
Team and time	Supervisor + workers who do the job + OHS adviser; one to three hours per job.
Key references	OSHA 3071 Job Hazard Analysis [R15]; ISO 45001:2018; national OHS (K3) regulation.

9.1 Key concepts

JHA (also called Job Safety Analysis, JSA) focuses on the relationship between the worker, the task, the tools and the work environment. Controls are chosen using the **hierarchy of controls**: elimination, substitution, engineering controls, administrative controls, and personal protective equipment (PPE) — in that order of preference. In an ANSP, JHA also has a service-safety angle: a maintenance task can itself remove a barrier (e.g. taking a radar out of service), so the JHA should reference the operational coordination required.

9.2 Procedure

1. Select and prioritise jobs: those with injury history, high severity potential, new or infrequent tasks.
2. Involve the workers; observe the job being done where possible.
3. Break the job into 5–12 sequential steps (“what is done”, not “how”).
4. For each step identify hazards: falls, electrical, RF radiation, moving parts, manual handling, heat, weather, vehicles, working alone.
5. Specify controls following the hierarchy; assign responsibility.
6. Review with the team, trial, then approve; review after any incident or change in the task.

9.3 Worked example — replacing an obstruction light on a radar tower

Step	Hazards	Controls (hierarchy)	Responsible
1. Coordinate outage and obtain permit to work	Radar removed without ATS awareness; RF exposure if radar left radiating	ATS coordination and NOTAM as required; radar transmitter isolated and locked out (LOTO); permit signed by radar engineer and TWR/APP supervisor	Site supervisor
2. Inspect climbing system and weather	Defective ladder/fall arrest; lightning; high wind	Pre-use inspection of fall-arrest rail; stop work if lightning within 10 km or wind above limit	Climber + supervisor
3. Climb tower with tools	Fall from height; dropped objects	Full-body harness, 100% tie-off; tools tethered; exclusion zone below	Climber
4. Isolate light circuit and replace lamp	Electric shock	Isolate at breaker, LOTO, test for dead; insulated tools	Technician
5. Descend, restore power, test light	Fall; energising while person on tower	Head count before de-isolation; permit closure sequence	Site supervisor
6. Return radar to service and hand back to ATS	Radar not restored; wrong configuration	Functional check; formal hand-back to supervisor; log entry	Radar engineer

9.4 Tips and pitfalls

- Too many or too few steps makes the JHA useless; a step is something a worker would recognise as a distinct part of the job.
- PPE is the last line of defence; a JHA whose controls are all PPE and “be careful” needs rework.
- Use the JHA as the basis for the pre-job briefing; a JHA in a file cabinet protects nobody.

10. FMEA / FMECA — Failure Mode and Effects (and Criticality) Analysis

At a glance	
Purpose	Bottom-up analysis of how each component or function of a system can fail, the effect of each failure mode on the service, how it is detected and compensated, and how critical it is.
When to use	Design and procurement of CNS/ATM systems, power and network architecture; maintenance programme design; checking that redundancy works as intended; input to FTA.
Inputs	System architecture and block diagrams; component list; reliability data (MTBF/failure rates); monitoring and switchover design.
Outputs	FMEA/FMECA worksheet; list of single points of failure; ranking (criticality or action priority); design and maintenance actions.
Team and time	Systems engineer, ATSEP, operational expert, safety assessor; days to weeks depending on system size.
Key references	IEC 60812:2018 Failure modes and effects analysis (FMEA and FMECA) [R16]; MIL-STD-1629A [R17]; SAE ARP4761A [R20].

10.1 Key concepts

For each item, the analyst lists its **failure modes** (e.g. no output, erroneous output, intermittent output, spurious output), their **causes**, the **local effect**, the **end effect** on the ATS, the **detection** method and the **compensating provisions**. FMECA adds a criticality measure. Two ranking approaches are common:

Risk Priority Number (RPN) = Severity × Occurrence × Detection, each rated 1–10. It is simple but mathematically weak (different combinations give the same RPN); IEC 60812:2018 therefore also describes alternatives such as criticality matrices and action priority tables.

Quantitative failure-mode criticality (MIL-STD-1629A): $C_m = \beta \cdot \alpha \cdot \lambda_p \cdot t$, where λ_p is the item failure rate, α the fraction of failures in this mode, β the conditional probability that the mode produces the end effect, and t the operating time. The item criticality is the sum of C_m over its modes with the same severity class.

10.2 Procedure

1. Define the system, boundaries, level of indenture and mission (e.g. provide air-ground voice on sector frequency, 24 h).
2. Draw the functional/reliability block diagram.
3. For each item list failure modes, causes, local and end effects.
4. Record detection means (BITE, monitoring, ATCO noticing) and compensating provisions (main/standby, backup radio).
5. Rate severity with the AirNav scale (end effect), occurrence and detection, or compute C_m .
6. Rank, identify single points of failure, recommend actions; re-rate after actions.

10.3 Worked example — sector VHF air-ground communication

Item	Failure mode	Cause	End effect on ATS	Detection	Compensation	S	O	D	RPN
Main VHF transmitter	No RF output	PA failure	Loss of main Tx on frequency	Radio monitoring alarm; no pilot reply	Auto switch to standby Tx	4	4	2	32
Main VHF transmitter	Stuck PTT (continuous carrier)	Relay / VCCS fault	Frequency blocked for all users	ATCO notices; timer alarm	PTT time-out; switch site	8	3	4	96

Item	Failure mode	Cause	End effect on ATS	Detection	Compensation	S	O	D	RPN
VCCS	Total loss	Software fault; power	Loss of all voice at CWP	Immediate	Backup radios at CWP	8	2	1	16
Leased line to remote site	Loss	Carrier outage	Loss of remote coverage (low levels)	Link alarm	Second path via VSAT	6	5	2	60
Main + standby receivers	Degraded sensitivity	Water ingress in antenna or cable	Weak/unreadable pilot calls	Late — pilot complaints	None	6	3	7	126

The highest RPN (126) is not the most severe failure: it is driven by poor detection. The recommended action is receiver-sensitivity monitoring (e.g. a site test transponder or periodic loop test) so that degradation is detected before pilots report it. The stuck-PTT mode (RPN 96, severity 8) shows why the severity rating should also be reviewed on its own: any mode with high severity deserves attention regardless of RPN.

10.4 Tips and pitfalls

- FMEA looks at single failures one at a time; it misses combinations and common causes — use FTA and common-cause analysis for those.
- Human error modes can be included (“ATCO selects wrong frequency”) but are better handled with HAZOP, STPA or task analysis.
- Keep S, O and D scales defined and published; undefined 1–10 scales make RPNs meaningless.

11. LOPA — Layer of Protection Analysis

At a glance	
Purpose	For a single cause–consequence scenario, estimate the frequency of the consequence after all independent protection layers (IPLs) have acted, and compare it with a tolerable frequency target.
When to use	When a scenario from HAZID/HAZOP/bowtie is rated high and the team needs to know whether the existing barriers are enough, or how much a new barrier would help. Bridges qualitative bowtie and full quantitative FTA/ETA.
Inputs	Scenario definition; initiating event frequency; enabling conditions and conditional modifiers; candidate IPLs with probability of failure on demand (PFD); tolerable frequency target.
Outputs	LOPA worksheet; mitigated event frequency; gap to target; required additional risk reduction.
Team and time	Safety assessor + ATCO + engineer; one to two hours per scenario once data is available.
Key references	CCPS, Layer of Protection Analysis: Simplified Process Risk Assessment (2001) [R18]; CCPS Guidelines for Initiating Events and Independent Protection Layers (2015) [R19]; IEC 61511.

11.1 Key concepts

LOPA originated in the process industry and maps naturally onto the ANSP view of barriers. The mitigated event frequency is:

$$f_C = f_{IE} \times \prod P_{CM} \times \prod PFD_i$$

where f_{IE} is the initiating event frequency, P_{CM} are enabling conditions or conditional modifiers (probability the situation needed for harm exists), and PFD_i is the probability that IPL i fails on demand. A safeguard may only be credited as an IPL if it is:

- **Independent** of the initiating event and of every other IPL credited in the same scenario;
- **Effective (specific)**: capable of preventing the consequence on its own, in time;
- **Dependable**: its PFD can be justified from data or accepted generic values;
- **Auditable**: its performance can be validated periodically.

Independence in ATM. Independence is the hardest criterion in ATM. Controller monitoring and STCA both depend on the same surveillance data path; if the initiating event involves corrupted surveillance, neither is independent of it. ACAS II uses its own interrogations and is independent of ground surveillance, but it depends on both aircraft having serviceable transponders. Apply LOPA only to scenarios where the assumed independence has been checked, and use FTA with common-cause analysis where it has not.

11.2 Procedure

1. Select a scenario (one initiating event → one consequence) from the bowtie/HAZOP.
2. Estimate f_{IE} from AirNav occurrence data (count ÷ exposure).
3. Identify enabling conditions and conditional modifiers (traffic present, geometry, time of day).
4. List safeguards; test each against the IPL criteria; assign PFDs (order-of-magnitude values: 10^{-1} for a trained human response with adequate time, 10^{-1} – 10^{-2} for a well-maintained automated alert, as justified).
5. Compute f_C and compare with the tolerable frequency; if above target, identify additional IPLs or reduce f_{IE} .
6. Document assumptions; hand residual risk to the hazard log.

11.3 Worked example — level bust leading to a potential mid-air collision

Element	Description	Value
Initiating event	Aircraft deviates from its cleared flight level (level bust) in sector X	0.5 per year
Conditional modifier	Another aircraft is in a conflicting position at the time	0.05
IPL 1	ATCO detects deviation on situation display and resolves the conflict in time	PFD 0.1
IPL 2	STCA alerts and ATCO resolves (credited only for deviations IPL 1 missed, assuming independent attention cue)	PFD 0.1
IPL 3	ACAS II resolution advisory generated and followed by both crews	PFD 0.1
Mitigated frequency	$0.5 \times 0.05 \times 0.1 \times 0.1 \times 0.1$	2.5×10^{-5} per year
Tolerable target (illustrative)	Set by AirNav for this outcome class in this sector	1×10^{-5} per year
Gap	Target not met by a factor of 2.5	—
Proposed new IPL	Selected flight level (SFL) mismatch alert: the ground system compares the Mode S downlinked selected level with the cleared level and alerts the ATCO before the aircraft leaves its level	PFD 0.1
Revised frequency	$2.5 \times 10^{-5} \times 0.1$	2.5×10^{-6} per year — target met

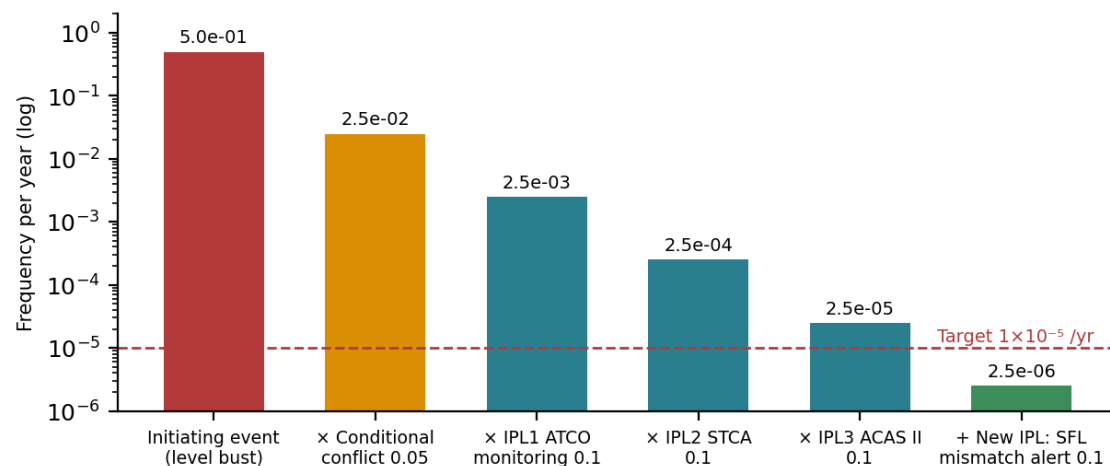


Figure 5. Frequency reduction through each layer in the level-bust scenario (log scale, illustrative values).

11.4 Tips and pitfalls

- Do not credit the same human twice (e.g. “ATCO monitoring” and “ATCO responds to STCA” are only partially independent; justify the credit or combine them).
- Use conservative, order-of-magnitude PFDs; LOPA is a screening tool, not a precise calculation.
- Procedures and training are not IPLs by themselves; they support the human IPL.

12. FHA — Functional Hazard Analysis

At a glance	
Purpose	Determine how safe a system or service needs to be: identify its functions, the ways each function can fail (failure conditions), the operational effects and severity of each, and set a safety objective (maximum tolerable frequency) for each failure condition.
When to use	At the start of a system or service change, before detailed design — the first step of the FHA → PSSA → SSA lifecycle used in EUROCONTROL SAM and in aircraft systems safety (ARP4761A).
Inputs	Functional description (what the system does, not how); operational environment; interfaces; risk classification scheme.
Outputs	Failure-condition list with effects, severity class and safety objectives; initial safety requirements; input to PSSA (FTA, FMEA, common-cause analysis).
Team and time	Safety assessor, operational experts (ATCO), system engineers; one to three workshops.
Key references	EUROCONTROL SAM — FHA chapter [R7]; SAE ARP4761A and ARP4754B [R20]; EUROCAE ED-78A for data-link services.

12.1 Key concepts

FHA works on **functions**, so it can start before architecture is chosen. For each function the team considers standard failure-condition types: **total loss**, **partial loss / degradation**, **erroneous or corrupted output** (detected and undetected), **unintended (spurious) function** and **delayed output**. Undetected erroneous output is usually the most severe, because nobody reacts to it. Each failure condition is assessed in its operational context (traffic, available fallbacks, exposure time) and assigned a severity class. The risk classification scheme then gives the maximum tolerable frequency — the **safety objective**. The later PSSA must show that the proposed architecture can meet each objective, and the SSA must show that the implemented system does.

12.2 Procedure

1. Describe the system functions and the operational environment; list assumptions.
2. For each function, identify failure conditions using the five types above.
3. For each failure condition, describe the operational effects, taking account of existing mitigations external to the system (procedures, other systems, ATCO and pilot actions).
4. Classify severity using Table 3.2 (worst credible effect).
5. Derive safety objectives from the risk classification scheme (Table 12.1); apportion if several failure conditions contribute to the same outcome.
6. Record initial safety requirements (e.g. “the system shall detect and flag position data with integrity below X within Y s”).
7. Hand objectives and requirements to the PSSA.

12.3 Worked example — ADS-B surveillance in an en-route sector

Function F1: “Provide the ATCO with the position, identity and pressure altitude of each ADS-B equipped aircraft in the sector, updated at least every N seconds.” Assumptions: sector also has procedural fallback; partial radar coverage above FL245.

ID	Failure condition	Operational effect	Severity	Safety objective (per operating hour)
FC-1	Total loss of ADS-B data for all aircraft (detected)	Revert to radar where available, procedural elsewhere; separation increases; workload peak during transition	C Major	$\leq 1 \times 10^{-5}$
FC-2	Loss of data for a single aircraft (detected)	Label coasts/drops; ATCO applies procedural separation to one aircraft	D Minor	$\leq 1 \times 10^{-3}$
FC-3	Undetected erroneous position of a single aircraft	ATCO separates on wrong position; possible loss of separation unnoticed until ACAS	B Hazardous	$\leq 1 \times 10^{-7}$
FC-4	Undetected erroneous position of multiple aircraft (ground processing error)	Several false positions; high collision risk	A Catastrophic	$\leq 1 \times 10^{-9}$
FC-5	Excessive latency of position updates	Displayed position lags actual; separation margins eroded in turns	C Major	$\leq 1 \times 10^{-5}$

Table 12.1 — The objectives follow from the AirNav matrix: a failure condition must at least reach the “tolerable (lower)” region, so severity A requires likelihood 1, B requires 2, C requires 3, D requires 4, using the quantitative bands of Table 3.1. Derived requirements include integrity monitoring (NIC/NACp filtering), comparison with radar where overlapping, and ground-processing software assurance proportionate to FC-4.

12.4 Tips and pitfalls

- Describe effects in the real operational context; the same failure can be Minor at night and Major at the evening peak — state the assumption.
- Do not take credit in the FHA for mitigations that are part of the system being assessed; those belong in the PSSA.
- Software and data failures belong in the FHA too: erroneous AIP data, wrong adaptation data, corrupted flight plans.

13. STPA — System-Theoretic Process Analysis

At a glance	
Purpose	Identify how the control structure of a socio-technical system — ATCOs, pilots, automation, procedures, management — can issue or fail to issue control actions that lead to hazards, including cases where no component has failed.
When to use	New concepts of operation, automation and decision-support tools, human-automation interaction, data link, complex interfaces between organisations. CAST (the accident-analysis variant) for investigations.
Inputs	Concept of operations; organisational and technical architecture; procedures; roles and responsibilities.
Outputs	Losses, hazards and system-level constraints; control structure model; unsafe control actions (UCAs); loss scenarios; requirements and controller constraints.
Team and time	Facilitator trained in STPA, operational and technical experts; several days for a medium-size system.
Key references	Leveson, Engineering a Safer World (2011) [R21]; Leveson & Thomas, STPA Handbook (2018) [R22]; Leveson et al., Assuring safety of NextGen procedures (ATM2013).

13.1 Key concepts

STPA is based on STAMP (Systems-Theoretic Accident Model and Processes), which treats safety as a control problem: accidents occur when the control structure fails to enforce safety constraints. Each controller (human or automated) has a **process model** — its belief about the state of the controlled process — and issues **control actions** based on it, receiving **feedback**. Many ATM occurrences are process-model problems: the ATCO believes an aircraft has left a level when it has not, or a pilot believes a clearance was for them. The STPA Handbook [R22] defines four steps:

1. **Define the purpose of the analysis:** losses, system-level hazards, system-level constraints.
2. **Model the control structure:** controllers, control actions and feedback (Figure 6).
3. **Identify unsafe control actions (UCAs)** using four types: (1) not providing the control action causes a hazard; (2) providing it causes a hazard; (3) providing it too early, too late or in the wrong order; (4) the action is stopped too soon or applied too long.
4. **Identify loss scenarios:** why UCAs could occur (inadequate feedback, flawed process model, conflicting controls, component failure) and why correct control actions could be improperly executed or not followed.

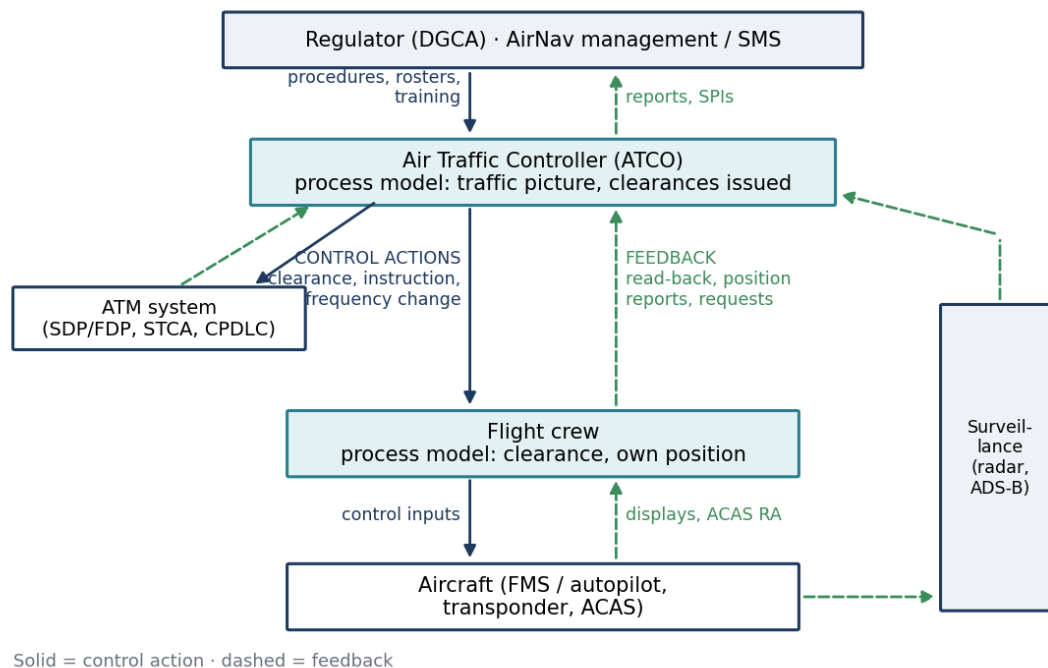


Figure 6. Simplified ATC control structure used in the STPA example.

13.2 Worked example — descent clearances in an approach sector

Losses: L-1 mid-air collision; L-2 controlled flight into terrain. **Hazards:** H-1 aircraft violate minimum separation [L-1]; H-2 aircraft descends below the minimum safe altitude for its position [L-2]. **System-level constraints:** SC-1 aircraft must maintain at least the minimum separation; SC-2 aircraft must remain at or above the applicable minimum altitude.

Control action	Not providing causes hazard	Providing causes hazard	Too early / too late / wrong order	Stopped too soon / applied too long
ATCO: descent clearance	UCA-1: ATCO does not issue descent when the aircraft must leave its level to avoid a conflict ahead [H-1]	UCA-2: ATCO clears descent through a level occupied by unseparated traffic [H-1]. UCA-3: ATCO clears descent below the minimum vectoring altitude [H-2]. UCA-4: descent is issued to a similar callsign [H-1]	UCA-5: descent issued too late to be established before the crossing point [H-1]. UCA-6: descent issued before the preceding aircraft has vacated the level [H-1]	Not applicable (discrete action)
Flight crew: execute descent	UCA-7: crew does not descend after a valid clearance while following traffic descends behind [H-1]	UCA-8: crew descends on a clearance meant for another aircraft [H-1]	UCA-9: crew starts descent before the clearance is complete (anticipation) [H-1]	UCA-10: descent continued through the cleared level (level bust) [H-1, H-2]. UCA-11: level-off too early, below the level assigned to crossing traffic [H-1]

Examples of loss scenarios

- **For UCA-6 (flawed process model):** the ATCO believes the preceding aircraft has left FL120 because its label shows a descending trend, but the Mode C update lags and the aircraft has been told to stop descent by the adjacent sector without coordination. Controls: coordination rules for level changes, SFL display, STCA.
- **For UCA-4 (inadequate feedback):** similar callsigns on frequency; the wrong aircraft reads back, the read-back is partially blocked and not challenged. Controls: callsign-similarity alerting, frequency congestion management, airline callsign de-confliction.
- **Conflicting controllers:** an ACAS RA and an ATC instruction disagree; the crew follows ATC. Constraint: crews shall follow the RA and ATCOs shall not issue contrary instructions once an RA is reported — a lesson reinforced by the 2002 Überlingen mid-air collision.
- **Organisational:** STCA inhibited in a sector because of nuisance alerts after an adaptation change; nobody owns the re-tuning. Constraint: changes to safety-net parameters require safety assessment and post-implementation monitoring.

13.3 Tips and pitfalls

- Write each UCA with its *context* (“when ...”); a UCA without context cannot be verified.
- Do not stop at UCAs — the requirements come from the loss scenarios.
- STPA does not need probabilities. Where the organisation needs a risk index, rate the resulting hazards with the matrix after the analysis.

14. FTA — Fault Tree Analysis

At a glance	
Purpose	Deductive, top-down analysis of the combinations of basic events (failures, errors, conditions) that lead to a specified undesired top event; qualitative (minimal cut sets) and quantitative (top-event probability, importance measures).
When to use	Demonstrating that an architecture meets a safety objective (PSSA/SSA); analysing loss of surveillance, communication, navigation or ATM functions; investigating technical occurrences; checking redundancy and common-cause vulnerability.
Inputs	Top event from FHA/bowtie; system architecture; FMEA; failure rates, repair times, test intervals; human error probabilities where relevant.
Outputs	Fault tree; minimal cut sets; top-event probability; importance ranking; design/maintenance actions.
Team and time	Reliability/safety engineer with system experts; days to weeks.
Key references	IEC 61025:2006 [R23]; NUREG-0492 Fault Tree Handbook (1981) [R24]; NASA Fault Tree Handbook with Aerospace Applications (2002) [R25].

14.1 Key concepts

Symbol / term	Meaning
Top event	The undesired state at the head of the tree, precisely defined (what, where, when, for how long).
OR gate	Output occurs if any input occurs. $P = 1 - \prod(1 - p_i) \approx \sum p_i$ for small p .
AND gate	Output occurs only if all inputs occur. $P = \prod p_i$ (inputs independent).
Basic event	Lowest level of resolution with a probability (component failure, human error, condition).
Undeveloped event	Not developed further for lack of information or relevance.
House event	A condition set to true/false to switch parts of the tree (e.g. maintenance mode).
Minimal cut set	Smallest set of basic events whose joint occurrence causes the top event. Single-event cut sets are single points of failure.
Common-cause failure (CCF)	Single cause failing redundant items together (shared power, software, network, maintenance error). Commonly modelled with a β -factor: $\lambda_{CCF} = \beta \cdot \lambda$.
Birnbaum importance	$I_B = P(\text{top} \mid \text{event occurs}) - P(\text{top} \mid \text{event does not occur})$: sensitivity of the top event to the event.
Fussell-Vesely importance	Fraction of top-event probability involving cut sets that contain the event.

14.2 Procedure

1. Define the top event and boundary conditions (initial state, operating mode, exposure time).
2. Develop the tree downward: for each event ask “what are the immediate, necessary and sufficient causes?”, choosing the gate type.
3. Continue to basic events with data; include human errors, CCFs and support systems (power, network, HVAC).
4. Obtain minimal cut sets by Boolean reduction (by software for large trees).
5. Quantify: assign probabilities (or rates $\times$ exposure, or unavailabilities for repairable items: $Q \approx \lambda \cdot \text{MTTR}$), compute top event and importance measures.
6. Compare with the safety objective; identify dominant cut sets and actions; perform sensitivity analysis.

14.3 Worked example — total loss of surveillance picture at an ACC sector

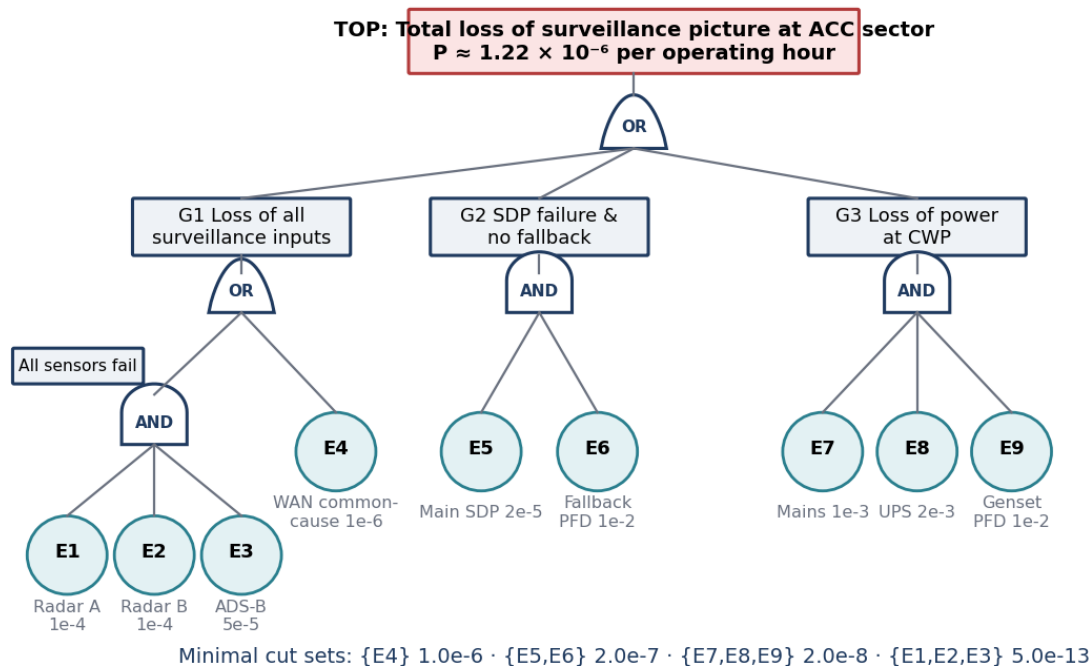


Figure 7. Fault tree for total loss of the surveillance picture (probabilities per operating hour or per demand, illustrative).

Basic event	Description	P	Birnbaum	Fussell-Vesely
E4	Common-cause loss of surveillance data network (single WAN)	$1.0\text{e-}6$	≈ 1.0	0.82
E5	Main surveillance data processor fails	$2.0\text{e-}5$	$1.0\text{e-}2$	0.16
E6	Fallback system fails to take over (PFD)	$1.0\text{e-}2$	$2.0\text{e-}5$	0.16
E7	Mains power failure	$1.0\text{e-}3$	$2.0\text{e-}5$	0.016
E8	UPS fails	$2.0\text{e-}3$	$1.0\text{e-}5$	0.016
E9	Generator fails to start (PFD)	$1.0\text{e-}2$	$2.0\text{e-}6$	0.016
E1-E3	Radar A, radar B, ADS-B fail independently	$1\text{e-}4, 1\text{e-}4, 5\text{e-}5$	$\leq 1\text{e-}8$	4×10^{-7}

The top-event probability is about 1.22×10^{-6} per operating hour, 82% of which comes from one single-event cut set: the shared data network. Triple sensor redundancy (E1-E3) contributes almost nothing, because a single network failure defeats it. If FHA had set an objective of 1×10^{-7} for this failure condition, the design would fail; the effective action is a diverse, physically separate data path (reducing E4), not another sensor. This is the typical lesson of FTA: redundancy is only as good as its independence.

14.4 Tips and pitfalls

- Mixing rates and probabilities is the most common numerical error. Convert everything to probability over the same exposure period, or to unavailability.
- Always search for CCFs: shared power, software, adaptation data, maintenance teams, building, network.
- The rare-event approximation (sum of cut sets) over-estimates slightly; it is acceptable when all cut-set probabilities are small.

15. Bio-mathematical fatigue modelling (three-process model)

At a glance	
Purpose	Predict the alertness (or sleepiness) of controllers and other safety-critical staff across a work schedule, so that rosters can be compared and high-fatigue periods identified and mitigated.
When to use	Designing or changing ATCO/ATSEP rosters; FRMS hazard identification; assessing overtime, extended shifts and quick returns; investigations where fatigue may have contributed.
Inputs	Planned (or actual) duty times; assumed or measured sleep times; time zone; model parameters.
Outputs	Alertness/KSS time-series per duty; time spent below thresholds; comparison between rosters; mitigation proposals.
Team and time	Fatigue/FRMS specialist, rostering staff, ATCO representatives; hours per roster once the model is set up.
Key references	Borbély (1982) two-process model [R26]; Åkerstedt & Folkard (1997) three-process model [R27]; Ingre et al. (2014) [R28]; ICAO Doc 9966 and the ICAO/CANSO/IFATCA Fatigue Management Guide for ATS Providers [R5, R6].

15.1 Background

ICAO Annex 11 requires States to establish regulations for managing fatigue in the provision of air traffic control services, either through prescriptive duty limits or through a Fatigue Risk Management System (FRMS) [R4, R5]. In both cases the ANSP must identify fatigue hazards in its rosters. Bio-mathematical models (BMMs) support this by predicting average group alertness from sleep-wake timing. They do not measure fatigue in an individual and must be combined with operational data (fatigue reports, subjective scales, sleep diaries, actigraphy, performance tests).

The **two-process model** (Borbély, 1982) explains sleep regulation by a homeostatic process *S* (sleep pressure builds during wake and dissipates during sleep) and a circadian process *C* (a roughly 24-hour rhythm). The **three-process model of alertness** (Åkerstedt and Folkard) adds process *W*, sleep inertia — the grogginess after waking — and expresses alertness on a scale that maps to the Karolinska Sleepiness Scale (KSS).

15.2 Model equations

Process	Equation (t in hours)	Parameters (as reported by Ingre et al. 2014 [R28])
<i>S</i> — homeostatic, awake	$S(t) = L_a + (S_w - L_a) \cdot e^{d \cdot t}$	Lower asymptote $L_a = 2.4$; $d = -0.0353$; S_w = level at waking
<i>S</i> — recovery during sleep	$S(t) = H_a - (H_a - S_s) \cdot e^{g \cdot t}$	Upper asymptote $H_a = 14.3$; $g \approx -0.381$; S_s = level at sleep onset. (The full model adds a “brake” that slows recovery above $S = 12.2$.)
<i>C</i> — circadian	$C(t) = C_a \cdot \cos(\pi(t - p)/12)$	Amplitude $C_a = 2.5$; acrophase $p = 16.8$ h (peak about 16:48)
<i>W</i> — sleep inertia	$W(t) = W_c \cdot e^{W_d \cdot t}$, t since waking	$W_c = -5.72$; $W_d = -1.51$
<i>U</i> — ultradian (optional)	12-hour component	Amplitude 0.5, mesor -0.5 (omitted in the example below)
Alertness and KSS	$A = S + C (+ U) + W$; $KSS \approx 10.6 - 0.6 \cdot A$	$KSS \geq 7$ (“sleepy”) corresponds to $A \approx 6$ or below

KSS	Verbal anchor (Åkerstedt & Gillberg, 1990 [R29])
1-3	Extremely alert · very alert · alert
4-6	Rather alert · neither alert nor sleepy · some signs of sleepiness
7	Sleepy, but no effort to keep awake
8-9	Sleepy, some effort to keep awake · very sleepy, great effort to keep awake, fighting sleep

15.3 Procedure

1. Collect the roster(s) to be assessed and define sleep assumptions (e.g. sleep 23:00–07:00 on normal nights; day sleep after night shifts), or use measured sleep.
2. Run the model for each roster pattern over enough days to reach a stable pattern (at least one full rotation).
3. Extract indicators for each duty: minimum alertness, maximum predicted KSS, time spent at $KSS \geq 7$, and the clock time of the low point.
4. Compare alternative rosters and mitigations (napping, break scheduling, shift start/end times, direction of rotation, number of consecutive nights).
5. Validate with operational data: KSS surveys at the start, middle and end of shifts, fatigue reports and, where possible, sleep diaries or actigraphy.
6. Record fatigue hazards and mitigations in the hazard log / FRMS.

15.4 Worked example — night shift with and without a pre-shift nap

An ATCO works a day shift (07:00–14:00) on day 1, sleeps 23:00–07:00, is off during day 2, then works a night shift 22:00–06:00, followed by day sleep 07:00–13:00. Two cases are modelled: no nap on day 2, and a 90-minute nap from 15:00 to 16:30.

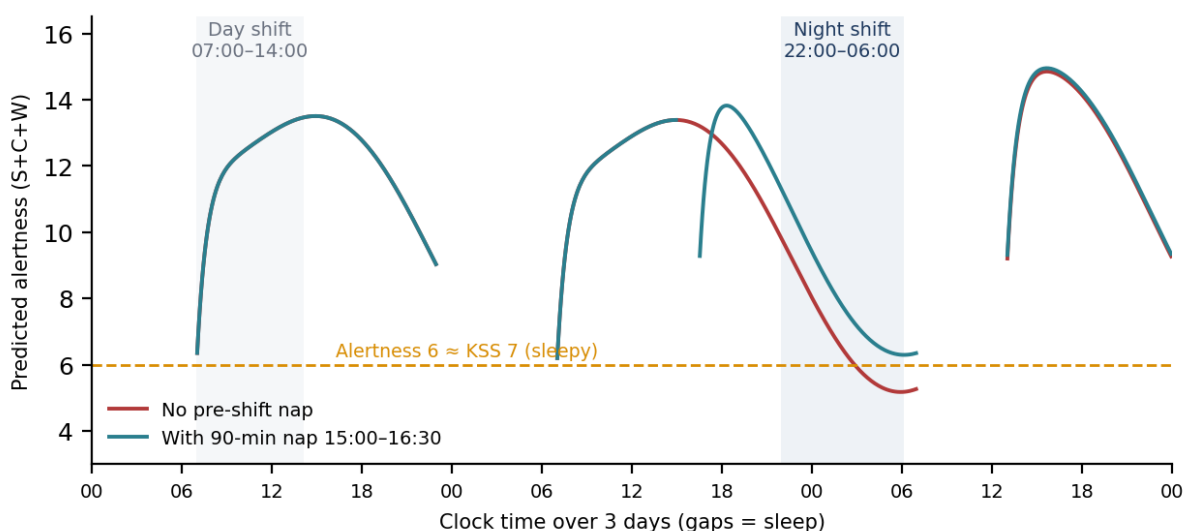


Figure 8. Three-process model prediction for the example roster ($S+C+W$, simplified model without the brake and ultradian terms).

Indicator (night shift 22:00–06:00)	No nap	90-min nap 15:00–16:30
Minimum predicted alertness	5.2 (at about 05:55)	6.3 (at about 05:55)
Maximum predicted KSS	7.5	6.8
Share of shift at $KSS \geq 7$	40% (about 3 h 10 min)	0%

The low point falls at the end of the night shift, where circadian trough and extended time awake coincide. The model suggests a pre-shift nap removes the predicted $KSS \geq 7$ period, but the last two hours remain close to the threshold, so further controls are still warranted: schedule the least demanding work and extra breaks between 03:00 and 06:00, protect in-shift rest/napping opportunities where permitted, avoid combining a night shift with an early start the following day, and limit consecutive night shifts.

15.5 Tips and pitfalls

- BMMs predict group averages for healthy adults. They do not account for workload, time on position, caffeine, individual differences (e.g. chronotype) or sleep disorders.
- Sleep opportunity is not sleep obtained: day sleep after a night shift is often shorter than planned. Use measured sleep where possible.
- Several commercial BMMs exist (e.g. SAFTE-FAST, the Boeing Alertness Model, FAID, CAS). Whichever model is used, document its version, parameters and validation, and never treat its threshold as a regulatory limit by itself.

16. FRAM — Functional Resonance Analysis Method

At a glance	
Purpose	Model how everyday work is actually done, as a set of coupled functions, and understand how normal performance variability can combine (resonate) to produce unexpected outcomes — both successes and failures.
When to use	Complex, tightly coupled operations where linear cause-effect models fall short: arrival management, runway operations, handovers, abnormal situations; investigation of occurrences with no obvious “failure”; Safety-II learning from what goes well.
Inputs	Interviews and observations of work-as-done; procedures (work-as-imagined); occurrence and normal-operations data.
Outputs	FRAM model (functions and couplings); description of variability; identified resonance paths; recommendations for monitoring and damping variability.
Team and time	Analyst trained in FRAM plus practitioners; days to weeks, mostly spent on data collection.
Key references	Hollnagel, FRAM (2012) [R30]; Hollnagel, Hounslow & Slater, FRAM Handbook (2018) [R31]; EUROCONTROL white papers on Safety-II and systems thinking [R32]; Tian & Caponecchia (2020) review of FRAM in aviation [R36].

16.1 Principles

- **Equivalence of successes and failures:** things that go right and things that go wrong happen in basically the same way.
- **Approximate adjustments:** because conditions are never fully specified, people and organisations continuously adjust their performance.
- **Emergence:** outcomes emerge from the combination of variability in many functions rather than from a single cause.
- **Functional resonance:** variability of several functions can combine and be amplified (or dampened) through the couplings between them.

16.2 The six aspects of a function

Aspect	Definition	Example for “Issue descent clearance”
Input (I)	What starts the function or is used/transformed by it	Arrival sequence; aircraft approaching top of descent
Output (O)	The result of the function	Descent clearance issued and read back
Precondition (P)	Condition that must be true before the function can start	Level below is free; coordination with next sector done
Resource (R)	What is needed or consumed while the function is carried out	Frequency time; ATCO attention; CWP
Control (C)	What supervises or regulates the function	LoA descent profiles; monitoring feedback; STCA
Time (T)	Temporal constraints	Must be issued before top of descent; sector workload window

16.3 Procedure (four steps)

1. **Identify and describe the functions** using the six aspects. Start from the function of interest and add upstream/downstream functions until the model is sufficient.
2. **Characterise variability** of each function's output. Technological functions are usually stable; human functions vary often and widely; organisational functions vary slowly but with large effect. Output variability is described for *timing* (too early, on time, too late, not at all) and *precision* (precise, acceptable, imprecise).
3. **Identify functional resonance**: trace how variability in one output affects the aspects of downstream functions it is coupled to, for a specific scenario (an instantiation of the model).
4. **Propose ways to monitor and manage variability**: dampen unwanted variability, amplify useful adjustments, add indicators.

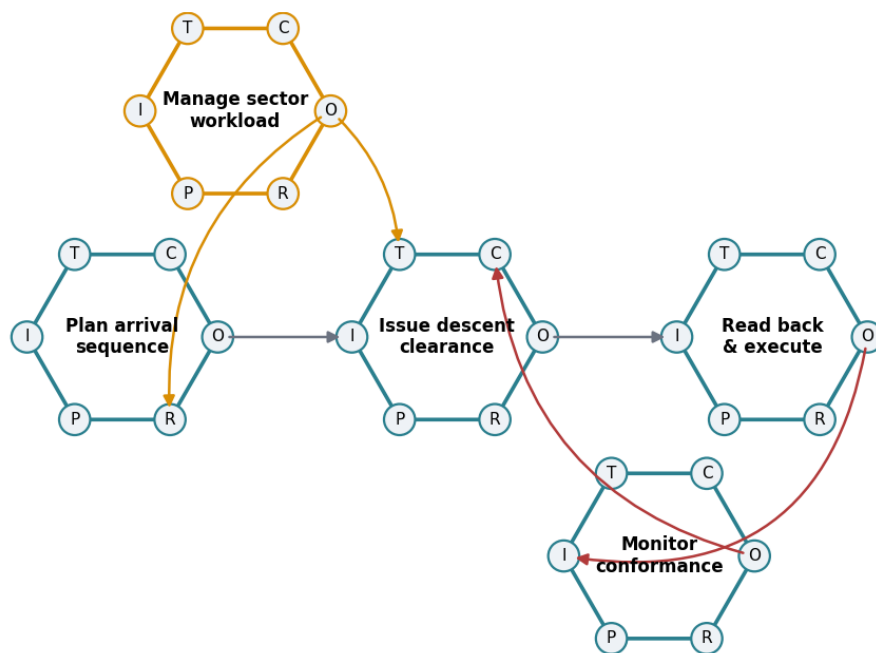


Figure 9. FRAM instantiation for descent clearances in a busy arrival sector. Red couplings: variability in read-back and monitoring feeding back into the clearance function; amber: workload acting on time and resources.

Function	Type	Likely output variability (busy peak)	Downstream effect	Management option
Manage sector workload	Organisational / human	Too late (split decision delayed)	Time and resources of all ATCO functions shrink	Traffic-load indicator triggers sector split earlier
Plan arrival sequence	Human + AMAN	Imprecise (sequence changes late)	Clearances revised; more RT	Freeze horizon for sequence
Issue descent clearance	Human	Too late; imprecise (abbreviated phraseology)	Read-back errors, steep descents	Standard phraseology refresher; CPDLC for non-urgent clearances
Read back & execute	Human (crew)	Imprecise (partial read-back, blocked)	Monitoring must catch the error	Hear-back emphasis; frequency-congestion SPI
Monitor conformance	Human + technology	Too late under workload	Level busts detected late	SFL display and mismatch alert

16.4 Tips and pitfalls

- FRAM needs good data on work-as-done: plan observations and interviews, not just procedure reading.
- Keep the model focused; a hundred-function model is rarely useful. The free FRAM Model Visualiser (FMV) helps build and share models.
- FRAM does not produce a risk index. Where one is needed, transfer the identified hazards to the matrix after the analysis.

17. BBN — Bayesian Belief Networks

At a glance	
Purpose	Represent causal relationships between uncertain factors as a directed acyclic graph with conditional probability tables, so that risk can be predicted from known conditions and causes can be inferred from observed outcomes.
When to use	Integrated risk models combining technical, human and organisational factors; linking precursors/SPIs to accident risk; updating risk estimates as new data arrive; diagnostic reasoning in investigations.
Inputs	Causal structure (from bowties, FTA, expert elicitation); occurrence and exposure data; structured expert judgement for missing probabilities.
Outputs	Network model; marginal probabilities of outcomes; posterior probabilities given evidence; sensitivity ranking of factors.
Team and time	Risk analyst with BBN skills, domain experts, data analyst; weeks for a first useful model.
Key references	Fenton & Neil (2018) [R33]; Ale et al. (2009), the CATS model for air transport safety [R34]; Pearl (1988) [R35].

17.1 Key concepts

Each node is a variable with discrete states (e.g. fatigue: yes/no). Arcs point from cause to effect. Each node holds a conditional probability table (CPT) giving its distribution for every combination of parent states. The joint probability factorises as $P(X_1, \dots, X_n) = \prod P(X_i \mid \text{parents}(X_i))$. Two kinds of reasoning follow:

Predictive (cause → effect): what is the probability of loss of separation if the ATCO is fatigued and workload is high? **Diagnostic** (effect → cause, using Bayes' theorem $P(A|B) = P(B|A) \cdot P(A) / P(B)$): given that a loss of separation occurred, how likely is it that fatigue was present?

17.2 Procedure

1. Define the purpose and the outcome node(s); keep the model as small as the question allows.
2. Build the causal structure with experts, reusing bowtie threats and barriers or FTA structure; check that the graph is acyclic.
3. Define node states precisely ("fatigued" = $KSS \geq 7$, for example).
4. Populate CPTs from data where available and by structured expert elicitation otherwise; document the source of every table.
5. Validate: check marginal outputs against observed rates, test extreme scenarios with experts, run sensitivity analysis.
6. Use the model for what-if analysis and update CPTs as data accumulate.

17.3 Worked example — coordination error and loss of separation at transfer of control

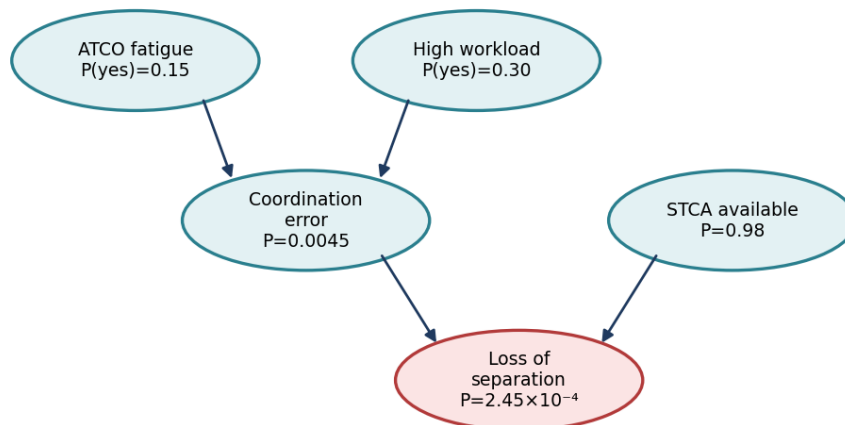


Figure 10. BBN for loss of separation following a coordination error at transfer of control (probabilities per transfer, illustrative).

Node	Conditional probability table
Fatigue (F)	$P(F = \text{yes}) = 0.15$
Workload (W)	$P(W = \text{high}) = 0.30$
Coordination error (E)	$P(E F, W): F \text{ yes \& } W \text{ high } 0.020; F \text{ yes \& } W \text{ normal } 0.008; F \text{ no \& } W \text{ high } 0.006; F \text{ no \& } W \text{ normal } 0.002$
STCA available (S)	$P(S = \text{yes}) = 0.98$
Loss of separation (L)	$P(L E, S): \text{error \& STCA } 0.05; \text{error \& no STCA } 0.30; \text{no error: } 0 \text{ (other causes outside this model)}$

Query	Result
$P(\text{coordination error})$	0.0045 per transfer
$P(\text{loss of separation})$	2.45×10^{-4} per transfer
$P(\text{LoS} \text{fatigue} = \text{yes})$ vs $P(\text{LoS} \text{fatigue} = \text{no})$	6.4×10^{-4} vs 1.8×10^{-4} (≈ 3.6 times higher)
$P(\text{LoS} \text{fatigue} = \text{yes}, \text{workload} = \text{high})$	1.1×10^{-3}
Diagnostic: $P(\text{fatigue} \text{LoS occurred})$	0.39 (prior 0.15)
Diagnostic: $P(\text{high workload} \text{LoS occurred})$	0.54 (prior 0.30)
Diagnostic: $P(\text{STCA unavailable} \text{LoS occurred})$	0.11 (prior 0.02)

The diagnostic results show why a BBN is useful for learning from occurrences: after a loss of separation the posterior probability of fatigue rises from 15% to 39%, so fatigue should be checked systematically in investigations of such events. The predictive results give a quantitative argument for fatigue and workload controls (e.g. earlier sector splitting), and the model can be updated as AirNav collects data.

17.4 Tips and pitfalls

- CPT size grows exponentially with the number of parents; keep to three or fewer parents per node, or use parametric forms (noisy-OR, ranked nodes).
- Expert-elicited probabilities are prone to overconfidence; use structured elicitation (calibration questions, multiple experts) and record it.
- Software: GeNIe/SMILE, Netica, AgenaRisk, Hugin, and the open-source Python library pgmpy all support model building and inference.

18. Integrating results and keeping them alive

18.1 From methods to one risk picture

Each method produces hazards, causes, controls and sometimes numbers. They must be brought together in one place — the hazard log — using one vocabulary and one risk classification scheme, otherwise the organisation ends up with twelve disconnected spreadsheets. The following cross-walk maps the terms used by each method onto the hazard log fields.

Hazard log field	Bowtie	HAZID / HAZOP	FMEA	LOPA	FHA	STPA	FTA	BBN
Hazard / top event	Hazard, top event	Hazard / deviation	End effect	Outcome	Failure condition	Hazard	Top event	Outcome node
Cause	Threat	Cause	Failure mode / cause	Initiating event	—	UCA, loss scenario	Cut set	Parent nodes
Control	Barrier	Safeguard	Detection, backup	IPL	Safety req.	Constraint	AND gate	Mitigating node
Likelihood	Qualitative	Matrix	O rating / C _m	Frequency	Objective	—	Probability	Probability

18.2 Minimum hazard log content

- Unique hazard ID, title and description; affected unit(s), system(s) and operation.
- Source assessment(s) and method(s); date; facilitator and participants.
- Causes/threats and consequences; existing controls with owner and verification status.
- Initial and residual risk index, with the basis of each judgement.
- Additional controls/actions with owner, due date and status; safety requirements and how they are verified.
- Acceptance decision, authority and date; review date; links to SPIs and to related occurrences.

18.3 Safety Assessment Report structure

1. Summary and conclusion (is the change acceptably safe, under which conditions?).
2. Description of the change, scope, boundaries and assumptions.
3. Methods used and why; participants.
4. Hazards identified and their assessment (reference to hazard log).
5. Safety objectives and requirements; evidence that they are or will be met.
6. Residual risk and acceptance; outstanding actions and conditions of implementation.
7. Post-implementation monitoring plan (SPIs, review date).
8. Annexes: worksheets, models, calculations, data sources.

18.4 Monitoring and review

Assessments rest on assumptions. After implementation, AirNav shall monitor that the assumptions hold and the controls work: link critical barriers and safety requirements to SPIs, review occurrence reports against the relevant bowties and models, and re-open an assessment when an assumption is invalidated, when an SPI crosses its alert level, or at the review date set in the hazard log.

Appendix A. Abbreviations

Abbrev.	Meaning	Abbrev.	Meaning
ACAS	Airborne collision avoidance system	HAZID	Hazard identification study
ACC	Area control centre	HAZOP	Hazard and operability study
ADS-B	Automatic dependent surveillance – broadcast	IPL	Independent protection layer
AIDC	ATS inter-facility data communication	JHA	Job hazard analysis
ALARP	As low as reasonably practicable	KSS	Karolinska sleepiness scale
AMAN	Arrival manager	LOPA	Layer of protection analysis
ANSP	Air navigation service provider	LoS	Loss of separation
APP	Approach control	LOTO	Lock-out / tag-out
A-SMGCS	Advanced surface movement guidance and control system	MSA / MVA	Minimum sector / vectoring altitude
ATCO	Air traffic controller	PFD	Probability of failure on demand
ATSEP	Air traffic safety electronics personnel	PSSA / SSA	Preliminary / system safety assessment
BBN	Bayesian belief network	RIMCAS	Runway incursion monitoring and collision avoidance system
CAST	Causal analysis based on STAMP	RPN	Risk priority number
CCF	Common-cause failure	SDP / FDP	Surveillance / flight data processing
CFIT	Controlled flight into terrain	SFL	Selected flight level
CPDLC	Controller-pilot data link communications	SMS	Safety management system
CPT	Conditional probability table	SPI	Safety performance indicator
CWP	Controller working position	SRM	Safety risk management
FHA	Functional hazard analysis	STCA	Short-term conflict alert
FMEA / FMECA	Failure mode and effects (and criticality) analysis	STPA / STAMP	System-theoretic process analysis / accident model and processes
FRAM	Functional resonance analysis method	UCA	Unsafe control action
FRMS	Fatigue risk management system	VCCS	Voice communication control system
FTA	Fault tree analysis		

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